The Impact of Product Uniqueness in AI Recommendation Systems on User Acceptance Willingness

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Abstract

Product uniqueness in AI recommendation systems significantly impacts consumer perceptions and behaviors. However, existing research on how product uniqueness in AI recommendations influences consumer acceptance is relatively scarce. Grounded in self-determination theory and empowerment theory, this study adopts a field and between-subjects experimental design to explore the internal mechanisms and boundary conditions of how product uniqueness in AI recommendations affects consumer acceptance. The findings reveal that: (1) the higher the product uniqueness recommended by AI, the higher the consumer's acceptance of the recommendation; (2) the need for uniqueness moderates the relationship between product uniqueness and consumer acceptance of AI recommendations; (3) perceived expertise plays a moderated mediation role in the relationship between product uniqueness and consumer acceptance of AI recommendations. These findings offer valuable insights into the fields of AI and marketing, while also assisting companies and e-commerce platforms in effectively utilizing AI recommendations to enhance recommendation efficiency and strengthen marketing capabilities.

CCS Concepts

• **Information systems, Information retrieval, Retrieval tasks and goals, Business intelligence**;

Keywords

AI recommendation, Product uniqueness, Algorithm transparency, Perceived autonomy, Acceptance willingness

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With the rapid development of internet technologies and the widespread adoption of artificial intelligence (AI), users frequently face challenges such as information overload and decision fatigue in online environments. To address these issues, AI recommendation systems have been extensively implemented in various domains, including e-commerce, social media, news platforms, and streaming services [1]. For example, Taobao's recommendation system suggests products of potential interest to users, while Tencent Video recommends content based on users' viewing preferences [2]. Similarly, platforms such as Toutiao and TikTok deliver news articles and short videos that are highly aligned with user interests [3]. While these systems effectively cater to general user preferences, they often fall short in recommending niche or unique products, limiting their reach to potential consumers. As consumer demands become increasingly personalized, there is a growing preference for niche products that resonate with individual identities, as opposed to mainstream products [4]. This shift underscores the need for current recommendation systems to better identify and fulfill personalized consumer preferences. While advancements in data accumulation and computational power offer the potential for more precise and personalized recommendations, AI systems that rely predominantly on similarity algorithms may overlook the consumer's desire for uniqueness, leading to lower acceptance intentions and decreased satisfaction. Thus, this study focuses on product uniqueness within AI recommendation systems, exploring its influence on consumer acceptance intentions and offering insights for optimizing AI-driven marketing strategies.

1 Literature Review

1.1 AI Recommendations

The rapid development of the internet has expanded consumers' shopping channels, but the vast array of product choices has also led to the problem of choice overload. When consumers are faced with too many options, they may experience decision paralysis or delay in making decisions, which negatively impacts satisfaction and overall shopping experience [5]. Recommendation systems help consumers reduce the cost of information search and processing. Compared to traditional recommendation methods, the application of artificial intelligence (AI) has significantly improved the quality of recommendations, enhanced user experience, while also boosting sales for online retailers [6]. AI-powered recommendations excel in both professionalism and accuracy, particularly in tasks that require structured knowledge but not creativity. As a result, AI recommendations have become deeply integrated into

various aspects of online shopping and have emerged as a critical area of research.

1.2 Product Uniqueness

Product uniqueness is typically defined as consumers' subjective perception of a product's distinctiveness in the market. According to Lee et al. (2020), product uniqueness arises from consumers' integration and processing of available product information (such as sales volume, scarcity, etc.) and their cognitive resources. By comparing the attributes of different products, consumers can form a judgment about the uniqueness of a product [7]. Chen et al. (2019) further pointed out that consumers' evaluation of product uniqueness includes two dimensions: the first is a comparison based on similar attributes of different products, which reflects the differences between products; the second is the identification of unique attributes of novel products, which demonstrates the product's exclusivity [8]. High levels of product uniqueness are often characterized by factors such as scarcity, visual appeal, premium pricing, or customization in the purchase process, which can draw more consumer attention and significantly influence preferences and purchasing behavior. In meeting consumers' need for self-expression, product uniqueness plays a crucial role [9]. Overall, product uniqueness significantly impacts consumers' cognition, preferences, and purchasing behavior. Future research should explore how intelligent recommendation systems can better meet consumers' demands for unique products to enhance their purchase intentions and satisfaction.

2 Theoretical Analysis and Research Hypotheses

2.1 The Impact of Product Uniqueness on Acceptance Intention

Product uniqueness refers to the perceived distinctiveness and differentiation of a product in the market. Existing research has shown that product uniqueness plays a crucial role in consumer decision-making processes. First, product uniqueness often implies exclusivity. The higher the uniqueness of a product, the more it provides consumers with a sense of exclusivity, thereby enhancing perceived utility [10]. Moreover, product uniqueness helps consumers achieve self-differentiation goals. When consumers seek to express their individuality and distinguish themselves from others through purchasing behavior, highly unique products are better suited to meet this need, thus driving purchase behavior [11]. In certain consumption contexts, this need for self-differentiation becomes a dominant factor. Additionally, displaying product uniqueness has social value. By showcasing the unique products they purchase, consumers can gain social recognition and psychological satisfaction [12]. Studies have shown that product uniqueness, as an added value, positively influences consumers' purchasing decisions and plays a key role in stimulating consumer interest [13]. Based on this, we propose that product uniqueness adds value to products, helping consumers express their identity, convey specific messages, and gain various benefits [14]. The higher the product uniqueness of items recommended by AI, the greater consumers' acceptance intention. Thus, we propose the following hypothesis:

H1: The higher the product uniqueness of AI-recommended products, the more likely consumers are to accept the AI recommendation.

2.2 The Mediating Role of Perceived Autonomy

Perceived autonomy refers to the subjective feeling that consumers experience when making independent choices and decisions without external interference. According to Self-Determination Theory (SDT), autonomy, competence, and relatedness are the three fundamental psychological needs of individuals [15]. Empowerment Theory describes the process of delegating tasks and responsibilities [16]. According to this theory, consumers can delegate certain decision-making tasks to artificial intelligence (AI). Research shows that when AI-recommended products possess high uniqueness, consumers are more likely to accept AI's decision-making assistance, as it is not perceived as an infringement on their autonomy [17]. Therefore, consumers are generally more inclined to rely on AI for decisions related to high-uniqueness products, while they prefer to maintain a higher degree of autonomy when choosing low-uniqueness products. Based on this, we propose the following hypothesis:

H2: Perceived autonomy mediates the relationship between product uniqueness in AI recommendations and consumers' acceptance intention.

2.3 The Moderating Role of Algorithm Transparency

Algorithm transparency plays a crucial role in enhancing consumers' perceived autonomy. For consumers, the ability to make informed and well-considered choices is central to the experience of perceived autonomy [18]. Increasing the transparency of AI recommendation systems is an important approach to enhancing consumers' sense of autonomy. Transparent algorithms can improve the credibility of recommendations, fostering greater trust and a stronger sense of autonomous choice. In contrast, a lack of transparency in recommendation algorithms can undermine perceived autonomy, confuse decision-making processes, and leave consumers feeling uncertain and passive when making purchase decisions.

Existing studies suggest that when consumers are exposed to non-transparent micro-targeted or invasive advertisements, they often feel that their privacy is being violated and tend to resist these ads, even making choices that go against their preferences to preserve their autonomy. Therefore, increasing algorithm transparency is considered an effective means of enhancing consumers' perceived autonomy. Based on this, the current study posits that when AI-recommended products possess high uniqueness, consumers are likely to experience a stronger sense of autonomy, which in turn increases their acceptance of the recommendations. Therefore, the following hypothesis is proposed:

H3: Algorithm transparency moderates the effect of AI-recommended product types on consumers' perceived autonomy and their acceptance intention of the recommended products.

In summary, this study integrates the three hypotheses into a theoretical research model, as shown in Figure 1.

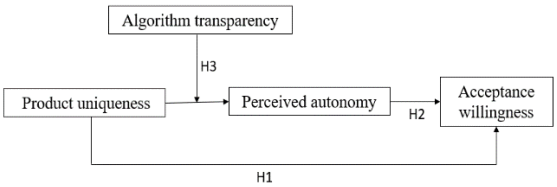

Figure 1: Conceptual Framework.

3 Experiment 1

3.1 Experiment 1

The primary objective of Experiment 1 is to verify the main effect, specifically that AI recommendations based on product uniqueness have a positive impact on consumers' acceptance intention. That is, the more unique the recommended product, the higher the consumer's willingness to accept the recommendation.

3.1.1 ***Experimental Design.*** Data for Experiment 1 was collected through the Credamo platform, with the experiment designed to simulate an online shopping experience. After excluding participants who failed the attention check, a total of 181 valid responses were obtained (62 male participants, 34.25%; M* = 25.29 years, SD = 4.69).

3.1.2 ***Experimental Procedure.*** Experiment 1 was conducted within the context of an online shopping experience and consisted of three parts: an introduction to the experiment, a description of the scenario and variable measurement, and demographic data collection. After browsing the products, participants were asked to respond to questions regarding their acceptance intention of the AI-recommended products, such as, "To what extent are you interested in the AI-recommended Bluetooth headset?" By clicking "next," participants were directed to a page measuring the moderating variable—need for uniqueness—along with other variables. Three attention check questions were embedded throughout the variable measurement pages to screen out inattentive participants. All items in the scales were measured using a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree). In the final section, participants were asked to provide basic demographic information, including gender, age, highest educational level, and online shopping experience.

3.1.3 ***Experimental Results.*** (1) Reliability Test:

A reliability test was conducted for both the need for uniqueness and acceptance intention scales. As shown in Table 1, the Cronbach's alpha values for each variable exceeded 0.85, indicating good reliability for all scales.

(2) Sample Overview

The demographic information collected from Experiment 1 includes gender, age, and highest educational attainment.

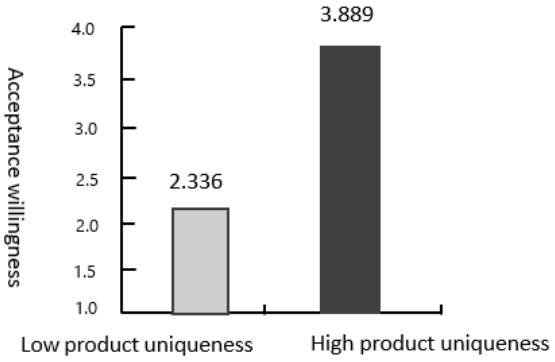

Figure 2: Influence of AI Product uniqueness on acceptance willingness.

In terms of gender distribution, males accounted for 34.25%, while females accounted for 65.75%, reflecting the higher proportion of females in online shopping.

Regarding age distribution, most participants were between 18 and 25 years old, representing 58.56% of the sample. As for educational background, the sample was primarily composed of participants with undergraduate and master's degrees.

(3) Main Effect Test

A correlation analysis was conducted on the variables of product uniqueness, acceptance intention, and need for uniqueness. The results showed a significant positive correlation between product uniqueness and the acceptance intention of the recommendation, indicating that the more unique a product, the higher the consumer's acceptance intention. Since product uniqueness is a binary variable, independent sample t-tests were conducted to further examine the effect of product uniqueness on acceptance intention for the overall AI recommendation, as well as for specific products such as clothing and Bluetooth headsets.

The results are presented in Table 2.

Overall, As shown in Figure 2, there were significant differences in participants' acceptance intentions depending on product uniqueness. When AI recommended products with higher uniqueness, participants exhibited a higher acceptance intention compared to products with lower uniqueness ($M_{\text{high uniqueness}} = 3.889$, SD = 1.48; $M_{\text{low uniqueness}} = 2.336$, SD = 1.34; $t = -3.66$, $p < 0.001$). Therefore, the main effect of product uniqueness was significant, H1 is supported.

Table 1: Test results of scale reliability and validity.

Variable	Item	Reliability test	Validity test
Acceptance willingness	4	Cronbach's 0.902	KMO Bartlett's 0.816 0.000

Table 2: The effect of AI recommendation based on product uniqueness on acceptance intention.

Product	Mean (Standard deviation)Product uniqueness-Low product uniqueness-High	t(low-high)	Sig(double)
Clothing	3.48(1.53) 4.52(1.66)	-6.23	<0.001
Headset	2.33(1.34) 3.89(1.48)	-3.66	0.001
Total	3.63(1.34) 4.47(1.48)	-5.66	<0.001

Table 3: Test results of scale reliability and validity.

Variable	Item	Reliability test	Validity test
Acceptance willingness	4	Cronbach's	KMO Bartlett's
		0.916	0.825 0.000
Perceived autonomy	4	0.925	0.862 0.000

3.2 Experiment 2

The main objective of Experiment 2 was to examine the mediating role of perceived autonomy in the relationship between different AI-recommended product types and consumers' acceptance intentions (H2).

3.2.1 ***Experimental Design.*** In this experiment, 242 online participants were recruited and compensated accordingly. A between-subjects single-factor design was employed, where the factor was AI-recommended product uniqueness (high uniqueness vs. low uniqueness). The process and manipulations in Experiment 2 were same in Experiment 1. To enhance the validity of the experiment, product types were changed (low uniqueness: printer vs. high uniqueness: candy), and participants were randomly assigned to one of the two groups corresponding to the type of AI-recommended product. After reading the experimental materials, participants were asked to complete ratings for the acceptance intention variable and to rate the perceived autonomy variable.

3.2.2 ***Experimental Procedure.*** Perceived autonomy was measured using L. Wang and H. Zhang (2021) 7-point scale (1 = strongly disagree, 7 = strongly agree), which included four items [19], such as "The AI recommendation does not affect my free will." The reliability of the scale was 0.747. To rule out convenience as an alternative explanation, Duarte et al.'s 7-point scale was used to measure convenience, which included two items, such as "It is easy to obtain the information needed to make a purchase decision," with a reliability of 0.788. Lastly, participants provided demographic information.

3.2.3 ***Results and Discussion.*** (1) Reliability and Validity Analysis

Cronbach's alpha coefficient was used to analyze the reliability of the perceived autonomy and continuous use intention scales. The KMO values for both scales exceeded 0.8, and Bartlett's test of sphericity showed p-values of 0.000, indicating that the scales had adequate structural validity. The detailed results are shown in Table 3.

(2) Main effect test

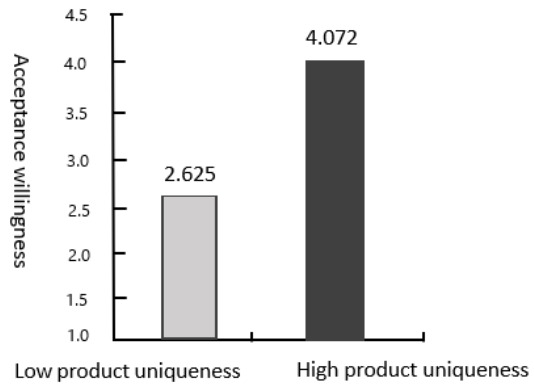

Figure 3: Influence of AI Product uniqueness on acceptance willingness.

Similar to experiment 1, ANOVA was used to test the uniqueness of artificial intelligence products. The results are shown in Table 4.

A one-way ANOVA was conducted to examine the effect of product uniqueness in AI recommendations on consumer acceptance intention. H1 was again validated.

As shown in Figure 3, the results indicated a significant main effect of product uniqueness in AI recommendations, F (167, 477) = 1572.442, p = 0.000. These results are consistent with H1, suggesting that consumers exhibit higher acceptance intention for AI-recommended products with high uniqueness (M = 4.072, SD = 1.16) compared to those with low uniqueness (M = 2.625, SD = 1.25).

(3) Mediation Effect Test

To test the mediation effect of the model, this study employed the Process plugin in SPSS and selected Model 4 for mediation effect analysis with 5,000 bootstrap samples. AI-recommended product type was used as the independent variable, perceived autonomy as the mediator, and acceptance intention as the dependent variable. As shown in Table 5, the results perceived autonomy had a significant mediation effect, with a mediation effect value of -0.0628, a standard error of 0.04, and a 95% confidence interval of

Table 4: ANOVA of AI Product uniqueness.

Variable		Ss	def.	Ms.	F	P
Acceptance willingness	Between Groups	167.477	1	167.477	1572.442	0.000
	Within Groups	11.183	105	0.107		
	Total	178.660	106			

Table 5: ANOVA of Perceived autonomy.

Effect type	Path	Effect	LLCI	ULCI
Direct effect	AI Product uniqueness- acceptance willingness	-0.062	-0.162	-0.001
Indirect effect	AI Product uniqueness - Perceived autonomy - Acceptance willingness	0.994	0.626	1.389

[-0.162, -0.001], which does not include 0. This indicates that perceived autonomy plays a significant mediating role. Therefore, H2 is supported.

3.3 Experiment 3

The primary aim of Experiment 3 was to examine the moderating role of algorithm transparency on both the main effect and the mediating effect in the research model.

3.3.1 ***Experimental Design.*** This experiment recruited 357 online participants who were compensated accordingly. A 2 (Product Uniqueness: High vs. Low) × 2 (Algorithm Transparency: High vs. Low) between-subjects mixed design was employed. Participants were randomly assigned to one of the four groups and exposed to the corresponding experimental stimuli. The product types used were the same as in Experiment 2 (office chairs and chocolates), but with the addition of an algorithm transparency manipulation.

3.3.2 ***Experimental Procedure.*** Participants were instructed to imagine that they were considering purchasing an office chair or chocolate online. They were browsing products on an e-commerce platform, and for the low algorithm transparency group, they were informed that the platform's AI recommendation system was suggesting certain office chairs or chocolates during their browsing session. Participants were required to complete ratings for acceptance intention and perceived autonomy variables. At the end of the experiment, participants provided demographic information, including gender, age, education level, and income, which were also used as control variables for statistical analysis.

3.3.3 ***Experimental Results.*** (1) Reliability Test

A reliability test was conducted for algorithm transparency, acceptance intention, and perceived autonomy. Result shown the Cronbach's alpha values for each variable were all above 0.85, indicating good reliability for the scales.

(2) Main Effect Test

A two-way ANOVA was conducted to analyze the experimental data. As shown in Table 6, the results showed that both AI-recommended product uniqueness ($F(1,353) = 13.23$, $p < 0.001$) and algorithm transparency ($F(1,353) = 15.98$, $p < 0.001$) had significant effects on consumers' perceived autonomy. Additionally, the interaction between product uniqueness and algorithm transparency was also significant ($F(1,353) = 10.59$, $p = 0.001$).

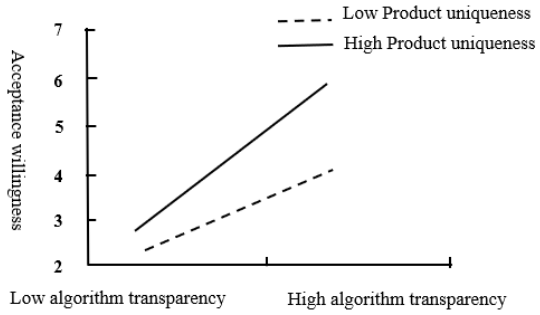

Figure 4: Simple slope test results for product uniqueness on acceptance at algorithmic transparency.

(3) Moderation Effect Test

To further validate H3, a moderated mediation effect test was conducted using the Process plugin in SPSS, selecting Model 8 with 5,000 bootstrap samples. Algorithm transparency was set as a categorical variable to examine its moderating role in the relationship between product uniqueness and consumers' acceptance intention. Table 7 presents the results.

Table 7 provides the direction of the moderation effect, allowing us to determine how the influence of product uniqueness on consumers' acceptance intention differs under varying levels of algorithm transparency. Based on the results from Table 7, Figure 4 was created. As shown in Table 7 and Figure 4, when algorithm transparency is high, consumers exhibit stronger acceptance intentions for products with high uniqueness compared to those with low uniqueness. This indicates that algorithm transparency moderates the relationship between product uniqueness and acceptance intention, thus supporting H3.

Table 6: ANOVA of Acceptance willingness.

Factors	F-value	p-value	Mean	SD
AI Product uniqueness	13.23	<0.001	4,16	1.09
Algorithm transparency	15.98	<0.001	4.86	1.12
Perceived autonomy	10.59	0.001	3.89	2.61
Acceptance willingness	24.22	<0.001	5.53	1.06

Table 7: Bootstrap Test for Moderation Effect.

Variable	Acceptance willingness	
	ßSD	t
Intercept term	1.551** 0.582	2.667
AI Product uniqueness	0.567*** 0.115	4.924
Algorithm transparency	4.655*** 0.832	5.597
AI Product uniqueness *Algorithm transparency	-0.790*** 0.166	-4.887
R^2	0.166	
F	13.521***	

4 Conclusion

This study, through three online experiments, compared the effectiveness of AI in recommending products with varying characteristics and explored the underlying mechanisms and boundary conditions influencing consumer acceptance intentions. The findings from Experiment 1 revealed that in real-world settings, AI recommendations for high-uniqueness products, as opposed to low-uniqueness products, lead to higher consumer acceptance intentions. Subsequent experiments confirmed that this effect holds true across different product types. Experiment 2 compared the uniqueness of printers and candy, demonstrating that the difference in consumer acceptance intentions for products with varying levels of uniqueness only exists in the context of AI recommendations. However, AI-recommended high-uniqueness products had a significantly better effect than low-uniqueness products. The results of Experiment 3 provided evidence of the mediating mechanism underlying the impact of AI-recommended product uniqueness on consumer acceptance intention, showing that AI recommendations for high-uniqueness (vs. low-uniqueness) products significantly enhance consumers' perceived autonomy, thereby leading to higher acceptance intentions.

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